DAY 3

Fire, climate, and your own question

Three ways to map fire, the climate that drives it, and a project of your own.

2 hours · Fire + Climate + Mini-project

Labs 5 and 6

RECAP

Two days in, here is what you can do

DAY 1

- Cloud-free composite of anywhere
- NDVI greenness map
- Hectares per land cover class
- The PROMPT recipe

DAY 2

- Your own forest type map
- Honest accuracy assessment
- Carbon stock and CO_2e
- Reporting a range with sources

Every one of those came with the same warning attached: **the code running is not evidence that the answer is right**. Today that warning gets its strongest demonstration.

PART ONE

Three questions about fire

They need three different datasets. Using the wrong one is the classic mistake.

GET THIS RIGHT FIRST

Which dataset answers which question

YOUR QUESTION	DATASET	WHAT IT ACTUALLY MEASURES
Where is it burning right now?	`FIRMS`	A hot pixel at the moment the satellite passed over. Near real time.
How many hectares burned?	`MODIS/061/MCD64A1`	The burn scar, mapped monthly. **This is the one for area.**
How badly did it burn?	dNBR, computed by you	Severity, from Sentinel-2 before and after.

THE CLASSIC MISTAKE

Active fire detections are not burned area. One fire burning for a week is counted many times over. A fire that starts and finishes between two overpasses is not counted at all. Counting FIRMS pixels and calling it hectares is wrong in both directions at once.

LAB 5, PART ONE

How much burned in Chiang Mai

● LIVE — Lab 5

BURNED AREA · MCD64A1 · JAN–MAY

2023	73,039 ha
2024	67,128 ha
2025	**33,482 ha**

TWO SENSORS, SAME STORY

YEAR	FIRMS PIXELS	BURNED HA
2023	6,409	73,039
2024	6,494	67,128
2025	3,311	33,482

2025 was roughly **half** the season 2024 was — and two completely independent instruments agree on that.

Independent agreement is the best evidence you will get that a trend is real rather than an artefact of one sensor.

Now measure severity — and watch it go wrong

NBR = (NIR - SWIR2) / (NIR + SWIR2)

Fire kills vegetation and dries the soil: NIR falls, SWIR rises, NBR drops.

dNBR = NBR before - NBR after. Bigger drop, worse burn. This is standard, published, USGS-endorsed methodology.

So we compute it over Chiang Mai for the 2024 season, apply the standard thresholds, and add up the hectares.

THE RESULT

1,594,272

hectares "burned"
according to dNBR

67,128

hectares burned
according to MODIS

Twenty-four times too much. More than half the province supposedly burned. And the map looks completely convincing.

WHY

Leaf-off looks exactly like burnt

SPECTRALLY IDENTICAL

Forest that burned
NIR falls · SWIR rises · NBR drops

↓

dNBR is right
this really did burn

Deciduous forest in April
NIR falls · SWIR rises · NBR drops

↓

dNBR is wrong
it is just the dry season

Northern Thailand is full of **deciduous forest**. Between December and April those trees drop their leaves — and leaf-off is spectrally almost identical to burnt ground. **dNBR cannot tell them apart**.

THE FIX

Let each dataset answer the question it is good at

dNBR is good at **how badly**. It is bad at **whether**.

So let MCD64A1 answer "whether", and compute severity *only inside a burn scar an independent product confirms*.

```
severity(dnbr)
  .updateMask(burn_scar)   # the whole fix
```

Total now: **67,147 ha** against MODIS's **67,128 ha**. A 0.03% difference — just rounding. **Now you can trust the breakdown**.

BURN SEVERITY WITHIN THE CONFIRMED SCAR · 2024

Unburned	3,702 ha
Low	18,897 ha
Moderate-low	25,356 ha
Moderate-high	16,680 ha
High	**2,511 ha**

Only **4%** burned at high severity — typical of the surface fires that dominate Southeast Asian deciduous forest.

When two methods disagree by 24×, do not pick the prettier map.

Find out which assumption broke. That instinct is the most valuable thing in this course, and it transfers to every dataset you will ever touch.

PART TWO

The climate behind the fire season

Two global datasets, twenty lines, one chart that explains everything.

LAB 6

Why Chiang Mai burns in February to April

● LIVE — Lab 6

CHIANG MAI 2024 · CHIRPS RAINFALL

Jan	**2.8 mm**
Feb	5.1 mm
Mar	12.3 mm
Apr	30.1 mm
Aug	318.6 mm

20 mm

total rainfall
Jan–Mar

770 mm

total rainfall
Jul–Sep

A **38-fold** difference between seasons. The dry season in northern Thailand is not merely drier — it is close to absolutely dry.

Peak temperature is **April, 28.8 °C** — not June. The hottest month is the end of the dry season, before the monsoon arrives and cools everything down.

Hot, bone dry, and carpeted in dead leaves

Rainfall collapses in January. Temperature peaks in April. For three months the forest is hot, dry, and — if it is deciduous — covered in fallen leaves.

That is the fire season.

You measured its extent in Lab 5. Now you have the physical explanation on one page, built from two global datasets in about twenty lines of code.

THIS CHART IS THE DELIVERABLE

A district officer or a journalist understands it in five seconds. That is worth more than any map you will make this week.

The skill is not producing the chart. It is knowing **which two variables to put on it** — and that came from understanding the forest, not the software.

PART THREE

Your turn

One province. One question. One defensible number.

MINI-PROJECT

The brief

You have about **25 minutes**. Produce three things:

1. **One map** — with a legend, a title, and the boundary drawn on it.
2. **One number** — hectares, tonnes, or percent.
3. **One sentence** stating the number, the dataset, and the year.

Scope small and finish. Do not scope big and have nothing to show.

PICK A QUESTION — ALL REUSE A LAB

How much of my province is forest?	Lab 2
Deciduous vs evergreen?	Lab 3
How much carbon does it store?	Lab 4
How much burned, last 3 years?	Lab 5
Is the dry season getting drier?	Lab 6
How much forest lost since 2001?	Hansen

HOW TO WRITE IT UP

The sentence is the deliverable

WRITTEN PROPERLY

"In Nan province, forest covers ***1,033,493 ha*** *(84% of the province), according to* ***ESA WorldCover v200*** *for the year* ***2021****."*

THE SAME CLAIM, WRITTEN BADLY

"Nan is 84.4% forest."

The second version cannot be checked, cannot be reproduced, and stops being true the moment somebody uses a different product. It also has a decimal place it has not earned.

Before you present, run the sniff test one last time: Is the number in the units I claimed? Does the total stay under the province area? Would a second dataset roughly agree? Can I name the dataset and year without looking it up? Does the map have a legend?

What you actually learned this week

The tools

- Cloud-free composites, NDVI, land cover
- Supervised classification with honest accuracy
- Carbon stock and CO_2 equivalent
- Fire extent, severity, and climate

These will go out of date. The dataset versions already change every year.

The habits

- **Specify precisely** — the PROMPT recipe
- **Check the number** before believing the map
- **Cite the product and the year**, always
- **When two methods disagree, find out why** — do not pick the one you like

These do not go out of date.

> You came in with almost no remote sensing background. You are leaving able to produce a defensible forest statistic for any province in Thailand — and, more importantly, able to tell when one you have been handed is wrong.

LAST THING

Post-test, and where to go next

Post-test now — the same questions as Day 1. It measures how much the course moved you, not how clever you are. Answer honestly; a wrong answer here is more useful to us than a lucky guess.

Then, three minutes each

Show your map. Say your number. Say where it came from.

KEEP THESE

- **The dataset cheat sheet** — verified IDs and units
- **The prompt cookbook** — works on any assistant, any dataset
- **Your notebooks** — they run for any province, change one line

Next step, whenever you want it: bring your own boundary as a shapefile instead of using a province, and everything in these notebooks works unchanged.